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Project Report
On
E-Commerce Sales Dashboard using Power BI

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Github: https://github.com/simran-yadav20/E-Commerce-Sales-Dashboard-

Abstract

This project presents an interactive **E-Commerce Sales Dashboard** developed using Power BI to analyze and visualize business performance. The dashboard integrates multiple aspects of sales data, including revenue, profit, customer segments, product categories, and regional distribution. By utilizing data visualization techniques such as line charts, bar graphs, pie charts, and map visuals, the project provides clear insights into trends and patterns.
The analysis reveals key findings such as the dominance of the technology category in revenue generation, the popularity of standard shipping methods, and higher sales concentration in specific regions. Additionally, customer segmentation highlights the contribution of different user groups, enabling better understanding of customer behavior.
The dashboard is designed to support data-driven decision-making by offering real-time insights and interactive filtering options. Overall, this project demonstrates how business intelligence tools like Power BI can transform raw data into meaningful insights, helping organizations improve performance, optimize operations, and enhance strategic planning.

1. Introduction

In the modern digital era, e-commerce platforms generate massive volumes of data from customer transactions, product sales, and operational activities. Analyzing this data is essential for businesses to understand market trends, customer behavior, and overall performance. Business Intelligence (BI) tools like Power BI play a crucial role in transforming raw data into meaningful insights through interactive visualizations.

This project focuses on developing an E-Commerce Sales Dashboard using Power BI to analyze and visualize key business metrics. The dashboard enables stakeholders to monitor sales performance, identify profitable segments, and make data-driven decisions efficiently.

2. Objectives

The main objectives of this project are:

- To analyze overall sales and profit performance
- To understand customer segmentation and behavior
- To identify top-performing product categories and sub-categories
- To evaluate shipping modes and their impact
- To analyze regional sales distribution
- To provide actionable insights for business improvement

3. Dataset Description

The dataset used in this project is the Superstore dataset, which contains transactional data of an e-commerce business.

Key Attributes:

- Order ID
- Order Date
- Customer ID
- Customer Name
- Segment (Consumer, Corporate, Home Office)
- Category (Technology, Furniture, Office Supplies)
- Sub-Category
- Sales
- Profit
- Quantity
- Region and State
- Ship Mode

The dataset provides a comprehensive view of business operations and is suitable for analytical and visualization purposes.

4. Tools and Technologies Used

- Power BI Desktop
- CSV Dataset (Superstore)
- Power Query (for data cleaning)
- DAX (Data Analysis Expressions)

5. Data Preprocessing

Before visualization, data cleaning and preprocessing were performed using Power Query:

- Removed null and missing values
- Corrected data types (Date, Numeric, Text)
- Removed duplicate records
- Renamed columns for clarity
- Ensured consistency across fields

These steps ensured data accuracy and reliability for analysis.

6. Data Modeling

The dataset was structured into a single table model. Relationships were not required as the dataset was already well-structured. Key measures were created using DAX for dynamic calculations.

DAX Measures:

- Total Revenue = SUM(Sales)
- Total Profit = SUM(Profit)
- Total Orders = DISTINCTCOUNT(Order ID)
- Total Customers = DISTINCTCOUNT(Customer ID)

7. Dashboard Design

The dashboard was designed with a user-friendly and visually appealing layout consisting of multiple visual elements.

7.1 Key Performance Indicators (KPIs)

- Total Sales
- Total Profit
- Quantity Sold
- Average Delivery Days

7.2 Visualizations Used

- Line Chart: Monthly Sales and Profit Trends
- Bar Chart: Sales by Category and Sub-Category
- Donut Chart: Sales by Segment and Payment Mode
- Map Visualization: State-wise Sales Distribution
- Slicers: Region Filter

8. Analysis and Findings

8.1 Sales Performance

The dashboard shows that total sales amount to 23.62K, while total profit is 1.7K, indicating a positive but moderate profit margin.

8.2 Category Analysis

Technology is the leading category in terms of revenue generation, followed by Office Supplies, while Furniture contributes the least.

8.3 Sub-Category Analysis

Phones emerge as the top-performing sub-category, followed by Tables and Storage.

8.4 Customer Segment Analysis

The Home Office segment contributes the highest sales, while Consumer and Corporate segments show balanced performance.

8.5 Regional Analysis

The West region, particularly California, shows the highest concentration of sales, indicating regional dominance.

8.6 Shipping Mode Analysis

Standard Class shipping is the most preferred option among customers, suggesting cost sensitivity.

8.7 Sales Trend Analysis

Monthly trends reveal fluctuations in sales and profit, with peaks observed during mid-year months, indicating possible seasonal demand patterns.

9. Key Insights

- Technology category drives maximum revenue
- Sales peak during mid-year months
- West region dominates sales performance
- Phones are the most demanded products
- Standard shipping is widely preferred
- Home Office segment contributes significantly to revenue

10. Output:

Super Store Sales Dashboard
Quantity
1658
Average Days
108
Profit
19.4K
Sales
117.04K
Central
East
South
West
Sum of Sales and Sum of Profit by State
Monthly Profit by YoY
Year 2019 2020
10K
0K
January February March April May June July August September October November December
Sales by Category
Office Su... 48K
Technolo... 42K
Furniture 27K
Sales by Ship Mode
Standard Class 29K
First Class 9K
Same Day 4K
Second Class 3K
Monthly sales by YoY
Year 2019 2020
20K
0K
Sales by Sub Category
Machines 18K
Chairs 14K
Binders 13K
Sum of Sales by Payment Mode
Cards 29%
COD 37%
Online 34%
Sum of Sales by Segment
Home Office 23%
Consumer 49%
Corporate 28%

Super Store Sales Dashboard
Quantity
1008
Average Days
96
Profit
6.0K
Sales
77.55K
Central
East
South
West
Sum of Sales and Sum of Profit by State
Monthly Profit by YoY
Year 2019 2020
4K
2K
0K
-2K
-4K
January February March April May June July August September October November December
Sales by Category
Technolo... 31K
Furniture 24K
Office Su... 23K
Sales by Ship Mode
Second Class 8.6K
First Class 8.2K
Standard Class 7.4K
Same Day 6.0K
Monthly sales by YoY
Year 2019 2020
20K
10K
0K
Sales by Sub Category
Tables 10.6K
Machines 9.8K
Phones 9.7K
Sum of Sales by Payment Mode
Cards 11%
Online 34%
COD 55%
Sum of Sales by Segment
Home Office 18%
Corporate 25%
Consumer 57%

11. Conclusion

The E-Commerce Sales Dashboard developed using Power BI provides a comprehensive overview of business performance. It enables users to analyze key metrics, identify trends, and make informed decisions. The interactive nature of the dashboard enhances usability and allows dynamic data exploration.

This project demonstrates how data visualization tools can transform raw data into valuable insights, ultimately helping organizations improve efficiency and achieve strategic goals.

12. Future Scope

- Integration with real-time data sources
- Addition of predictive analytics using machine learning
- Inclusion of return rate and customer churn analysis
- Deployment of dashboard on Power BI Service for sharing

13. References

- Microsoft Power BI Documentation
- Kaggle Superstore Dataset
- Online tutorials and BI resources